

Zeyu Gao (高泽宇)

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Research Interests: *Humanoid Whole-Body Control & Motion Retargeting · World Models for Robot Learning · Trustworthy & Medical LLMs · Generative Models*

EDUCATION

Nanyang Technological University, Singapore Nov 2025 – Oct 2026 (expected)

M.Sc. in Applied Artificial Intelligence

- All-A record across all coursework to date.

Guizhou University, China Sep 2020 – Jun 2024

B.Eng. in Artificial Intelligence

- Ranked top 10 in the major; University-Level Merit Student (2021–2022); Second-Class Scholarship (2022–2023).

PUBLICATIONS & SUBMISSIONS

(† equal contribution; * corresponding author)

[1] **FreqAlign: Bidirectional Frequency Alignment for Mitigating Contextual Hallucinations in LLM-based Extractive QA**

Zeyu Gao, Yizhang Liu*

Knowledge-Based Systems (SCI Q1), 2026 · First author · DOI: [10.1016/j.knosys.2026.116003](https://doi.org/10.1016/j.knosys.2026.116003)

- Proposed a decoding-time bidirectional frequency-alignment calibration mechanism: an attention–frequency feature classifier judges each token's contextual relevance, amplifying context-relevant low-frequency entities and numerals via local frequency statistics while suppressing irrelevant high-frequency tokens via global frequency.
- Significantly outperforms strong baselines on six extractive QA benchmarks, reducing entity-substitution and numerical errors by 59% and 66%, respectively, with only +3.2% inference latency.

[2] **FineForge: Dual-Driven Reinforcement Learning for Fine-Grained Detail Generation in Diffusion Models**

Heyuan Gao†, Bangxun Tang†, **Zeyu Gao**

ACM International Conference on Multimedia (ACM MM 2026), to appear · Second author · Project page:

<https://gaohy63601-droid.github.io/fineforge/>

- Formalized text and logo distortion in customized generation as a high-frequency structural-fidelity problem; proposed a dual-driven RL framework coupling DIFT-localized edge-aware rewards with AW-LFA dense feature distillation.
- Operates only at training time with zero inference overhead; on DetailBench, improves OCR accuracy by 14.87% (absolute) and reduces DreamSim relative error by 32.1%.

[3] **Morphology-Agnostic Animation-to-Robot Motion Retargeting via SceneFlow-Conditioned Diffusion**

Zeyu Gao, Zeran Su, Peiran Liu, Zicheng Duan, Zelin Tao, Qiang Zhang, Jiahao Ma*

Under review at NeurIPS 2026 · First author · Project page: <https://flowop-submission.github.io/>

- Introduced the new MA-A2R task: given an animated-character video and the kinematic specification of a target humanoid robot, directly generate executable, physically plausible joint-level motion trajectories.
- Developed the FlowOp framework, which disentangles motion content from the source morphology via sceneFlow and generates executable robot trajectories through dual-branch visual/skeletal encoding, Skeleton-Anchored Attention, and a Rectified-Flow DiT.
- Constructed the AnimBot benchmark of 7,800+ clips; achieves a cross-morphology MPJPE of 5.62, a 5.8× improvement over MoCapAnything.

[4] **MedGuardMem: A Verifiable Dual-Memory Framework for Expert-Level Medical Question Answering**

Zeyu Gao, Heyuan Gao

Under review at EMNLP 2026 · First author

- Upgraded medical memory injection into a verifiable dual-memory framework: K-Memory performs in-model knowledge recall via RQ-KMeans semantic IDs, while E-Memory compiles risky outputs into TCSA decoding constraints, bootstrapped and updated through a verifier-in-the-loop.
- Qwen3-8B reaches 27.26% on MedXpertQA-Text with the contraindication rate reduced from 19.3% to 6.1%; the 27B version reaches 43.75%, setting a new open-source state of the art.

[5] **A Review on Automatic Context Engineering**

Xinyu Tong, **Zeyu Gao**, Yixuan Chen, Baosheng Yu, Yibing Zhan, Yongcheng Jing, Dacheng Tao

Under review at EMNLP 2026 · Second author

- First dedicated survey on Automatic Context Engineering (ACE), proposing a type-by-step taxonomy that maps three classes of optimizable context — Guiding, Evidence, and Tool — onto a Baseline / Candidate / Scoring / Selection / Updating optimization loop.
- Systematically reviews automatic context-optimization methods and identifies key open problems such as tool-dimension sparsity, memory privacy, and cross-lingual generalization.

RESEARCH EXPERIENCE

Robot Skill Learning in Generative Interactive Worlds · Research Project *Ongoing*

- Leverage world models and video-generation models to automatically construct and complete diverse robot task environments.
- Convert generated visual worlds into simulation scenes endowed with geometric, physical, and interactive properties, turning environments from merely watchable into fully operable.
- Drive robot imitation learning with generative human demonstrations, closing the loop across environment generation, simulation training, and real-world deployment.

Command-Conditioned Humanoid Whole-Body Control · Research Project *Ongoing*

- Authored a library of reference humanoid motions in a game engine — e.g., forward, backward, and crouched walking on stairs, and other locomotion skills — and converted them into trackable joint-space reference trajectories.
- Extended NVIDIA's SONIC whole-body-control foundation model with a dedicated command branch, building a command-conditioned teacher policy that exploits privileged observations (e.g., height maps and simulator states) to accurately track the authored reference trajectories.
- Distilled the privileged teacher into a student policy that operates on onboard observations such as depth, following the teacher–student paradigm toward deployment under realistic sensing.

ProResUNet-Based Brain Tumor Segmentation System · Project Lead (Software Copyright) *Jul 2023 – Sep 2023*

- Designed ProResUNet, an enhanced U-Net architecture with pre-activation residual blocks and feature-fusion bilinear upsampling, improving the recovery of tumor boundaries and fine-grained structures in medical images.
- Built an end-to-end automatic brain-tumor segmentation and visualization system covering data preprocessing, model inference, and result presentation; granted a national software copyright.

Tax Knowledge Graph Construction & Visualization System · SRT Project Lead *Mar 2022 – Mar 2023*

- Designed a “Taxpayer Knowledge Forest” that parses statutory provisions into multi-faceted trees of subjects, objects, actions, and items, with temporal-validity and consistency constraints to support tax-rule reasoning and complex semantic queries.
- Developed a Python library for automated tax computation and relief simulation, encapsulating tax calculation, policy-constraint validation, and visualized query workflows to speed up regulation parsing and estimation.
- Delivered an easy-to-use visual interface for convenient system operation.

INTERNSHIP EXPERIENCE

ANGEL Lab, Nanyang Technological University — Research Intern *Nov 2025 – Present*

- Research on trustworthy medical large language models, focusing on medical knowledge memorization, clinical-reasoning enhancement, and safety constraints for high-risk outputs.

Daqing Oilfield Information Technology Co., Ltd. — AI Application Intern *Jan 2024 – Jun 2024*

- Helped optimize face-recognition algorithms and productionize multimodal identity-authentication solutions; participated in business-scenario adaptation and effectiveness validation.
- Contributed to the modular abstraction and API design of an AI service platform, enabling rapid integration and reuse of algorithm components.

Neusoft Corporation — Deep Learning Intern *Jul 2023 – Sep 2023*

- Conducted research and engineering on deep learning for medical imaging; independently developed the software for an automatic brain-tumor recognition system.

SKILLS & CERTIFICATIONS

- **Programming:** Python, PyTorch, C/C++, MATLAB
- **Certificates:** CET-6 (score 489); Intermediate Software Designer, China Computer Software Professional Qualification (Nov 2022)
- **Software Copyright:** Automatic Brain-Tumor Segmentation & Visualization System (Sep 2023)
- **Honors:** Outstanding Volunteer Award (Oct 2021)